

SkySentinel AI - Technical Code Documentation

SkySentinel AI - Drone Security Analyst Agent ?

SkySentinel AI is an autonomous, real-time Drone Security Analyst Agent designed to process and analyze synchronised drone telemetry and optical multi-spectral video feeds. Built as a high-fidelity Python prototype, it features a state-of-the-art Streamlit UI containing automated rule-based alerting, spatial coordinate tracking, database index queries, and a custom RAG (Retrieval-Augmented Generation) LangChain chatbot.

The prototype showcases the future of docked security drones, enabling property owners to completely automate boundary surveillance and audit logs using simple natural language.

? Interactive Web Dashboard Preview

The application contains four powerful dashboards:

1. **DRONE: Live Mission Control:** Features live multi-spectral feeds with computer vision thermal HUD overlays, continuous telemetry dials (Battery, Altitude, Speed), a live spatial coordinate tracker, and an auto-scrolling flashing red security exceptions console.
2. **STATS: Patrol Data Index:** A relational database auditor listing full-text indexed frames and spatial logs with real-time download and export triggers.
3. **CHAT: AI Security Analyst Chat:** A conversational RAG auditor permitting users to query histories (e.g. *"Were there any people spotted near the loading docks after-hours?"*) backed by LangChain tools.
4. **REPORT: Executive Patrol Summary:** Synthesizes a formal one-sentence mission narrative detailing security findings and generating detailed patrol checkpoint timelines.

?? Installation & Setup

Ensure you have Python 3.12 installed on your system.

1. Clone & Navigate to Repository

2. Install Project Dependencies

3. Setup API Configurations (Optional)

To unlock standard LLM/VLM processing, set your API credentials as system environment variables, or enter them directly inside the dashboard sidebar:

Note: If no API keys are provided, the prototype seamlessly activates its high-fidelity keyword/logical simulation engines, ensuring 100% offline functionality out-of-the-box.

4. Run the Streamlit Dashboard

5. Execute the Automated Test Suite

Verify core database schemas, logical rules, and agent reasoning structures:

? System Architecture & Design Decisions

1. Event-Driven Sync Architecture

- Decision: We synchronize asynchronous video frame feeds and telemetry packets using timestamp alignments, forming a unified event packet stored as a single relation in our database.
- Benefit: Guarantees telemetry spatial coordinates are permanently bound to the visual frame descriptions, resolving mapping and positional data drift.

2. Embedded SQLite & FTS5 (Full-Text Search)

- Decision: Instead of relying on a heavy vector database (like Chroma or Pinecone) which adds network latency and installation friction, we leverage SQLite combined with its built-in FTS5 extension.
- Benefit: The database is self-contained and highly portable. We achieve sub-millisecond keyword matches on high-volume frame visual descriptions, serving as a reliable retrieval base for our RAG assistant.

3. Dual VLM Operation

- Decision: We implemented both a mock VLM narrative engine and a live API client interface supporting Google Gemini (gemini-1.5-flash) and OpenAI (gpt-4o-mini).
- Benefit: Perfect for zero-cost offline evaluations while remaining fully ready to process live video file uploads via OpenCV.

? AI Assisted Workflow Impact Report

This prototype was developed with pair-programming assistance from Antigravity (built by Google DeepMind).

Tools Leveraged:

- Antigravity AI Agent: Assisted with the architecture design, automatic SQLite trigger configuration, complex logic of the rules engine, and full test suite drafting.
- Gemini Imagen 3: Generated high-fidelity mock drone thermal camera security violation frames to represent night-vision optical feeds in the Streamlit application.

Workflow Acceleration:

- Development Time Reduced by 80%: Instantly generated clean, modular boilerplate schemas and rules, allowing developers to focus strictly on dashboard UX and LangChain tools integration.
- Flawless SQLite Triggers: Automatically generated SQLite FTS5 index update triggers, resolving full-text index synchronization out-of-the-box.